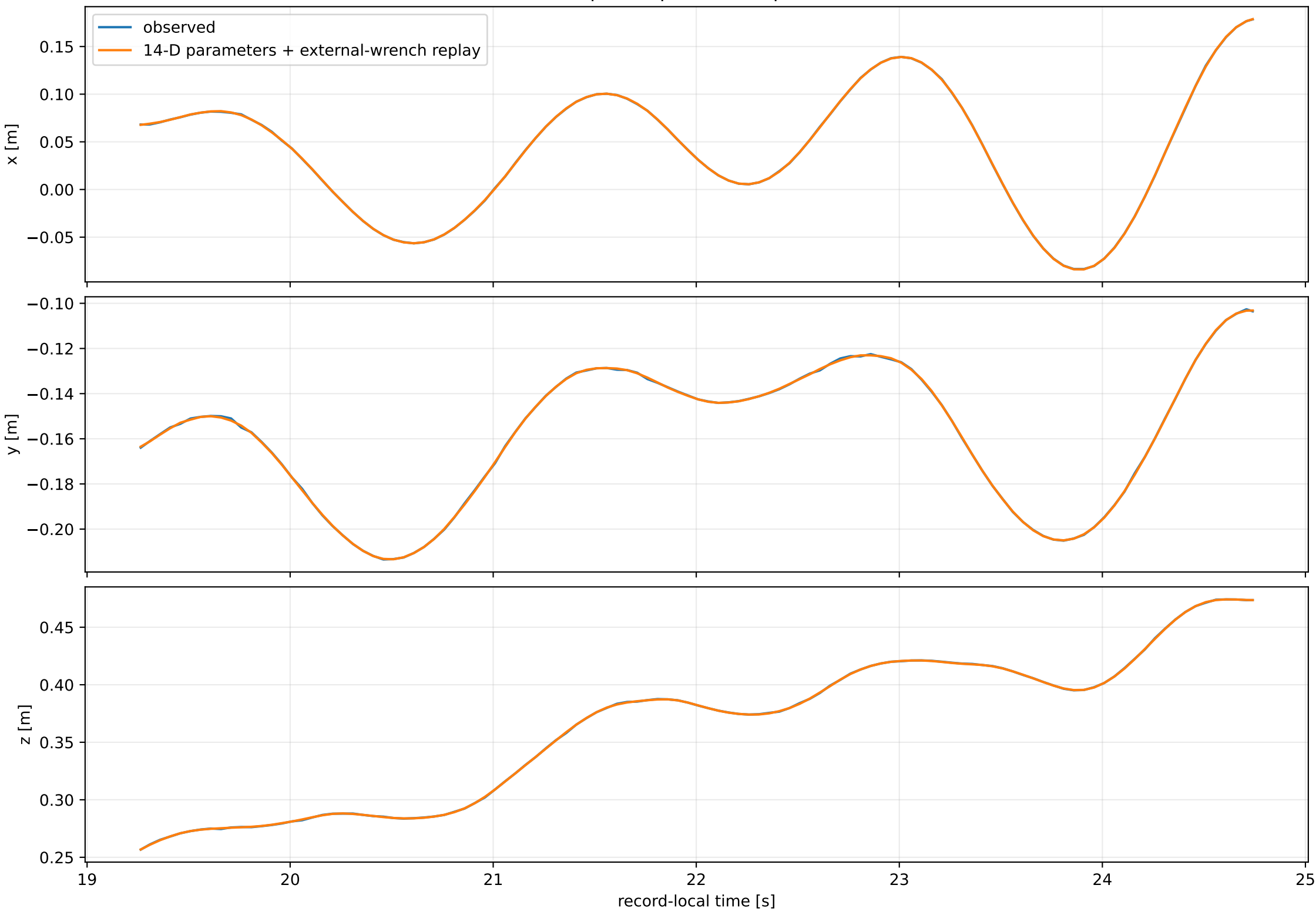
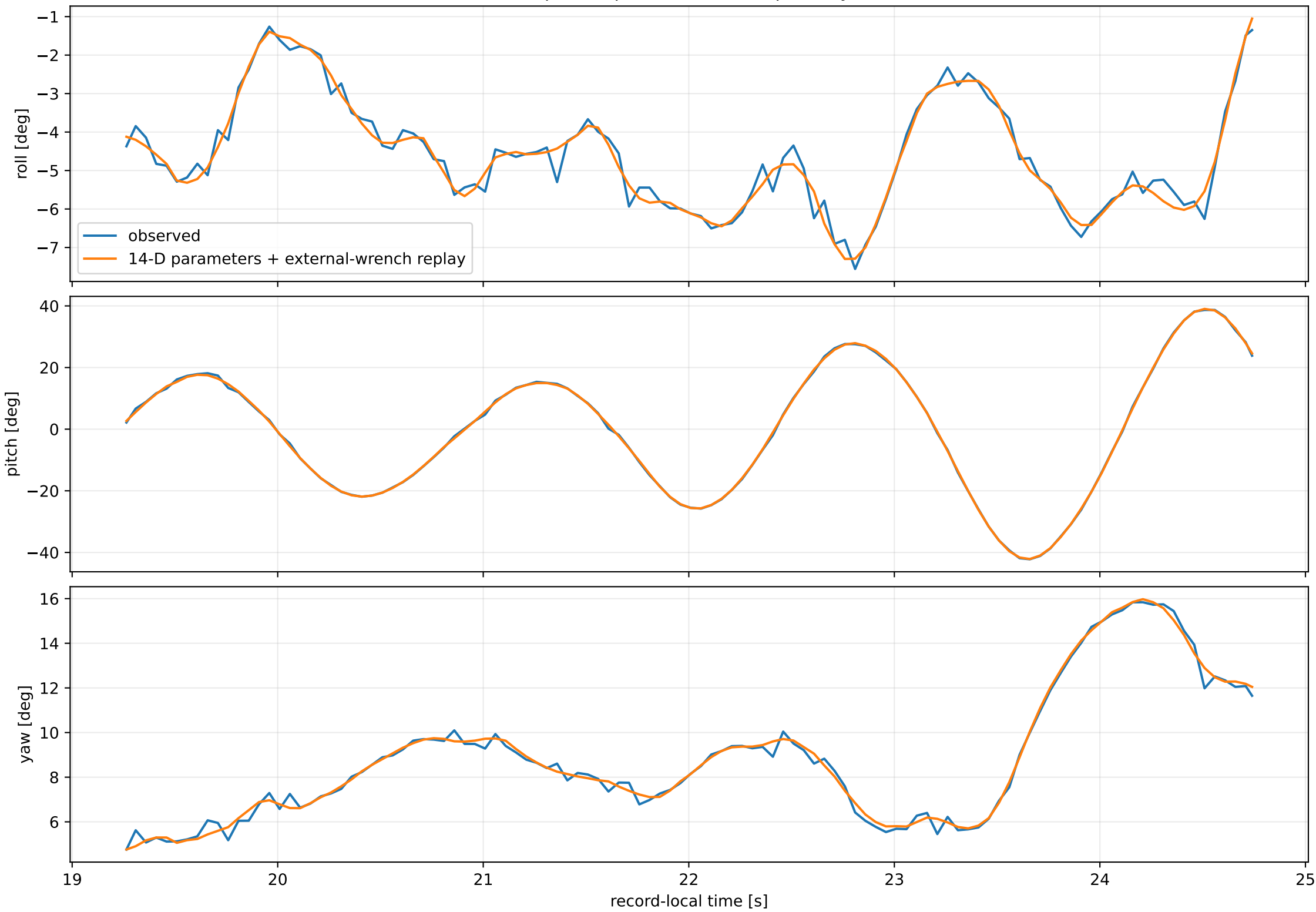


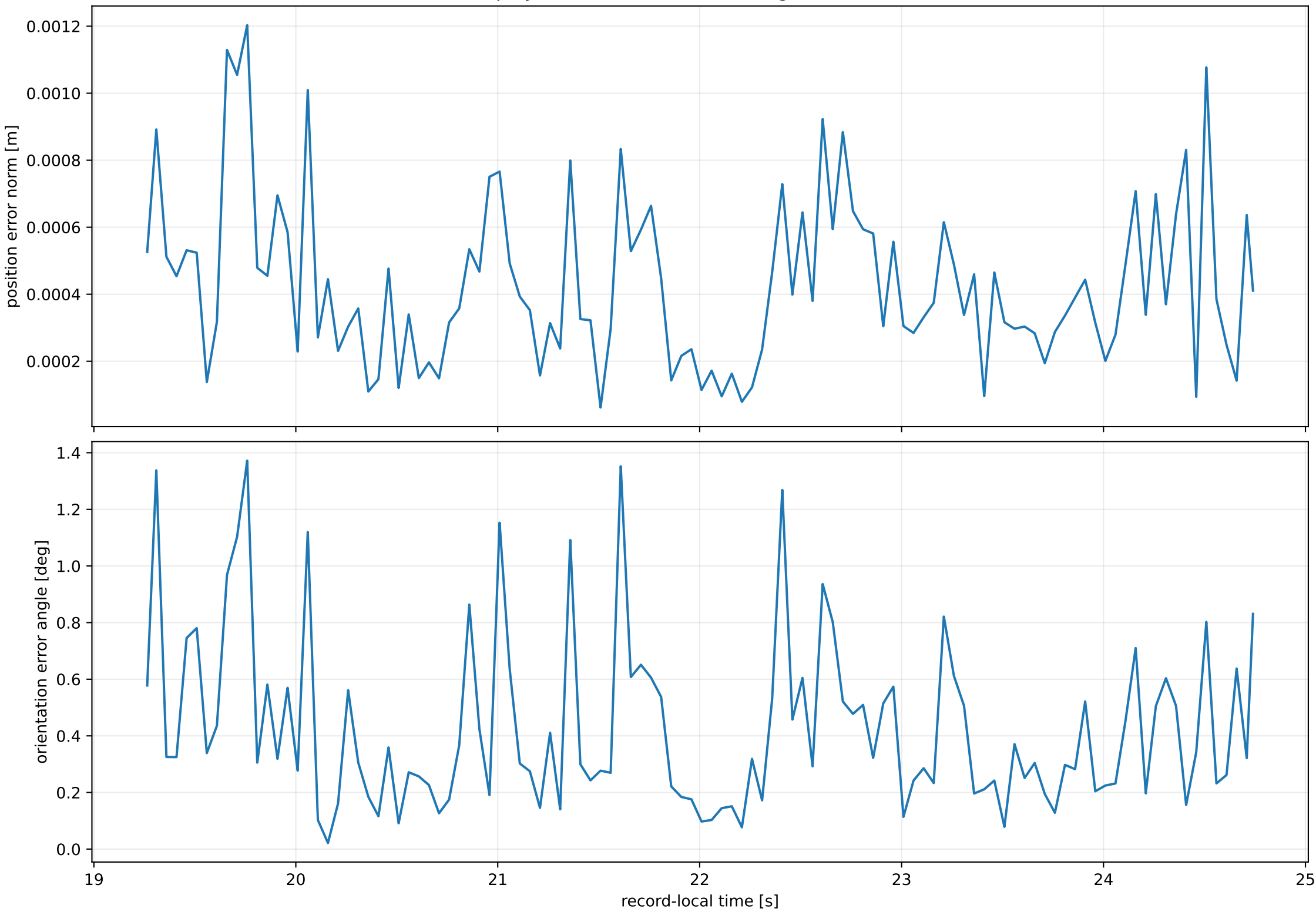
Fine pose reproduction: position



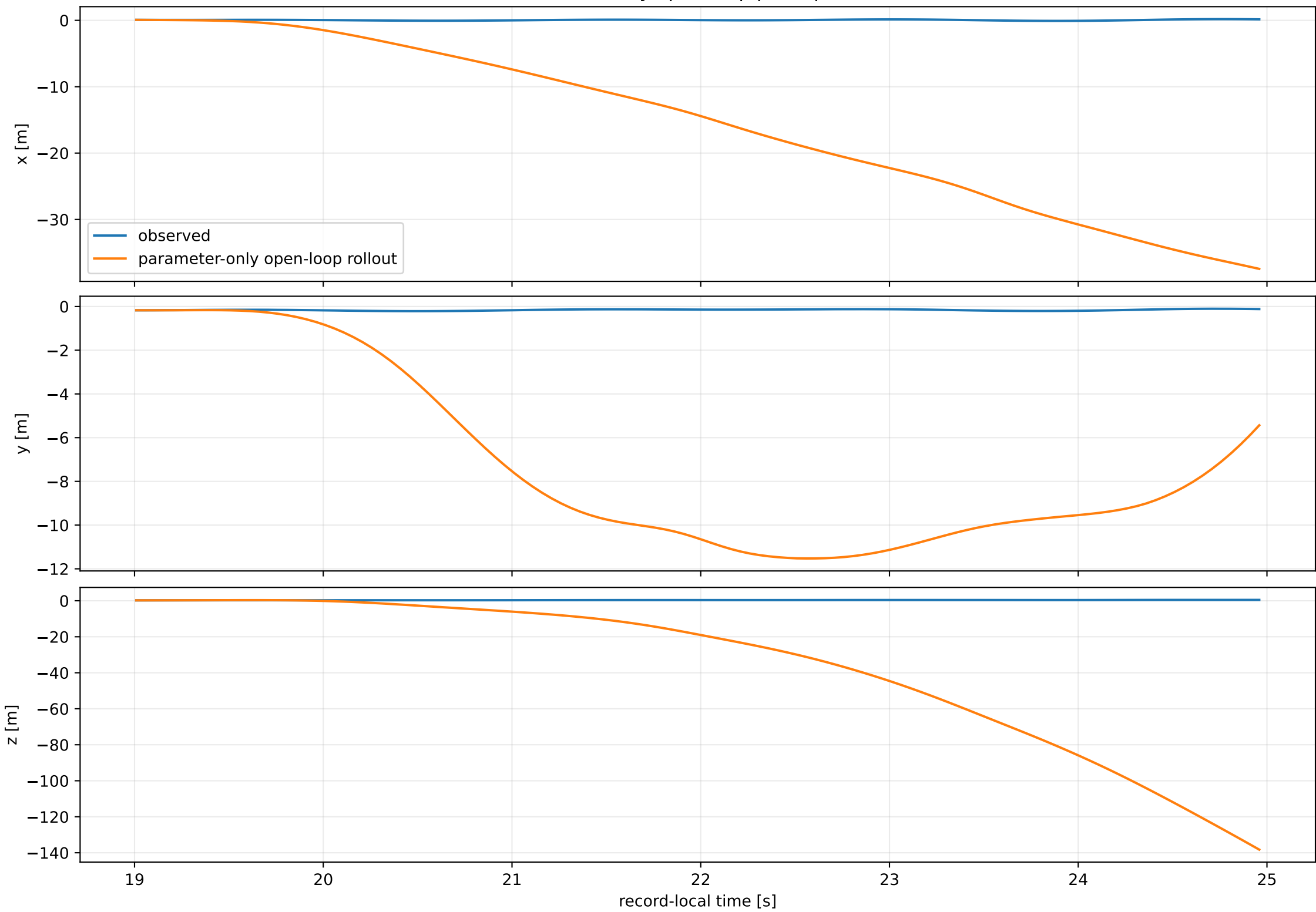
Fine pose reproduction: roll / pitch / yaw



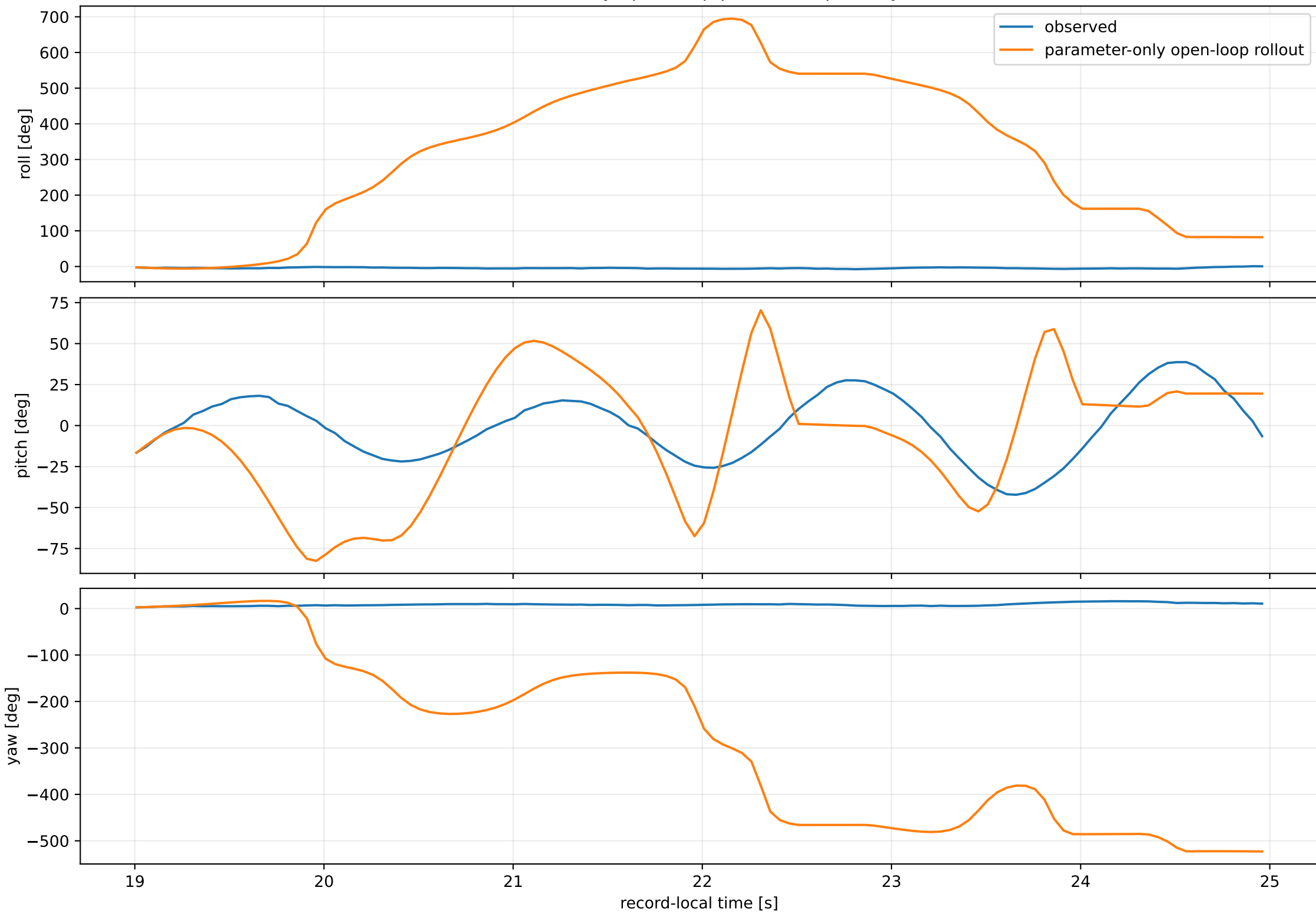
Fine replay error after the 14-D change — lower is better



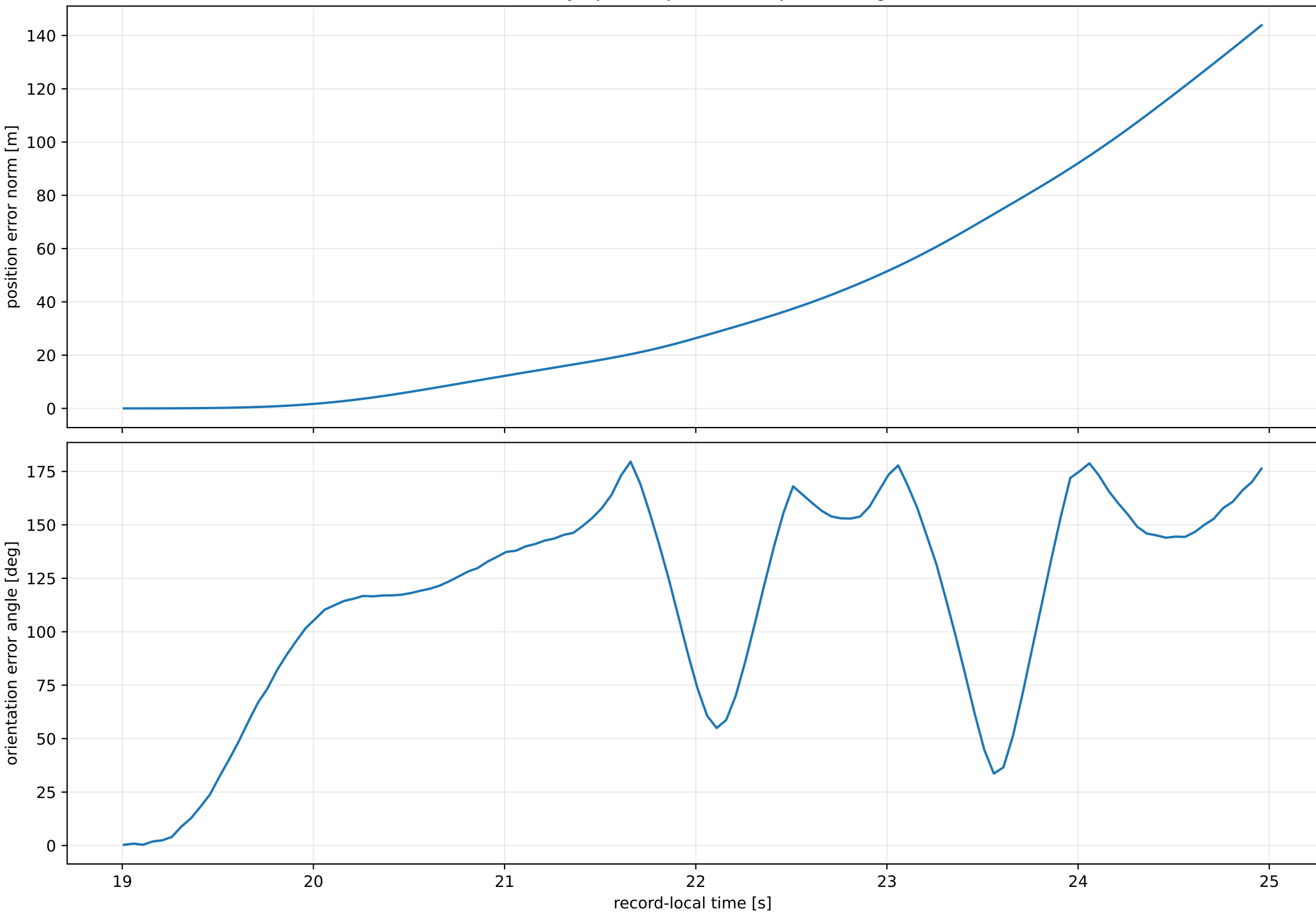
Parameter-only open-loop pose: position



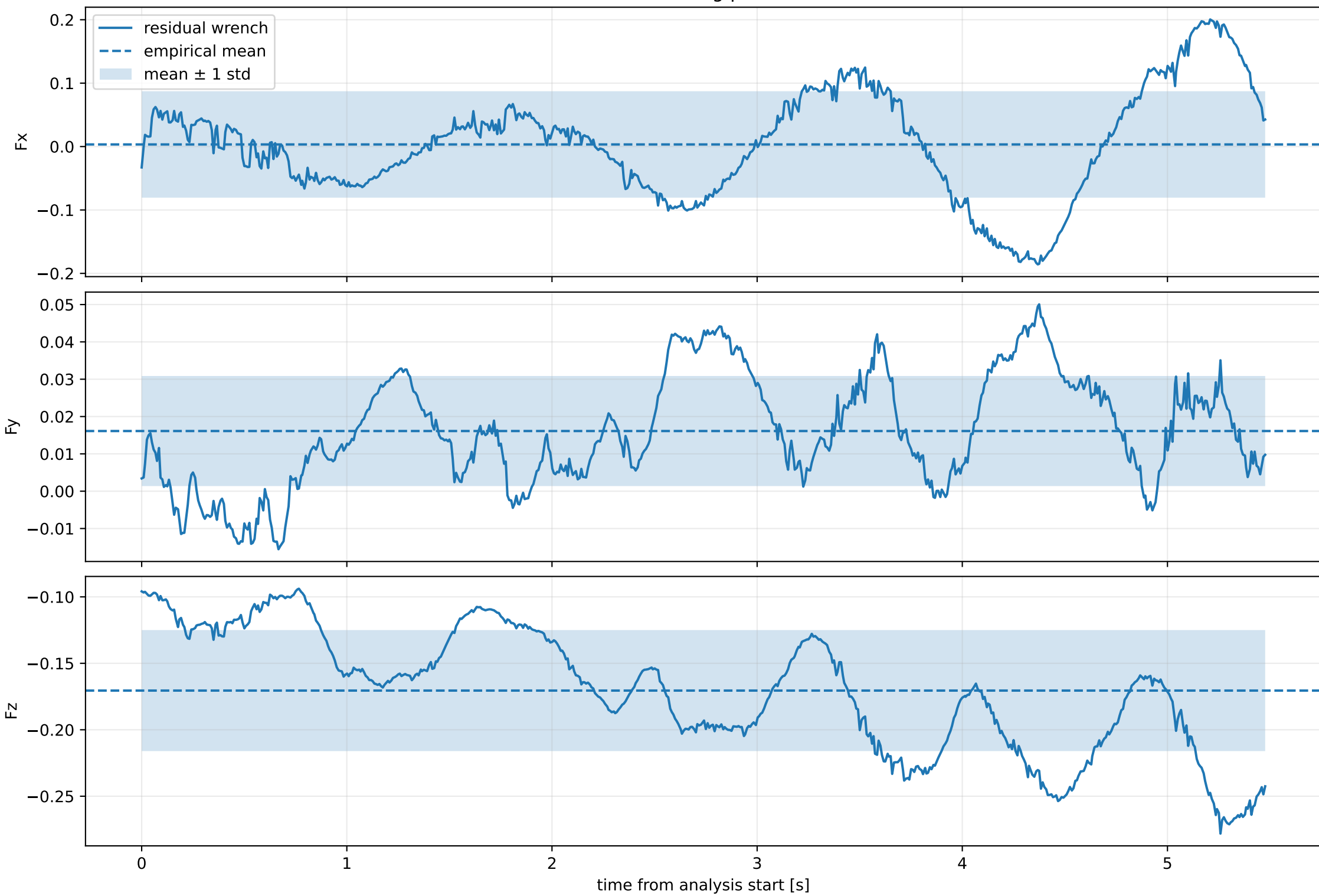
Parameter-only open-loop pose: roll / pitch / yaw



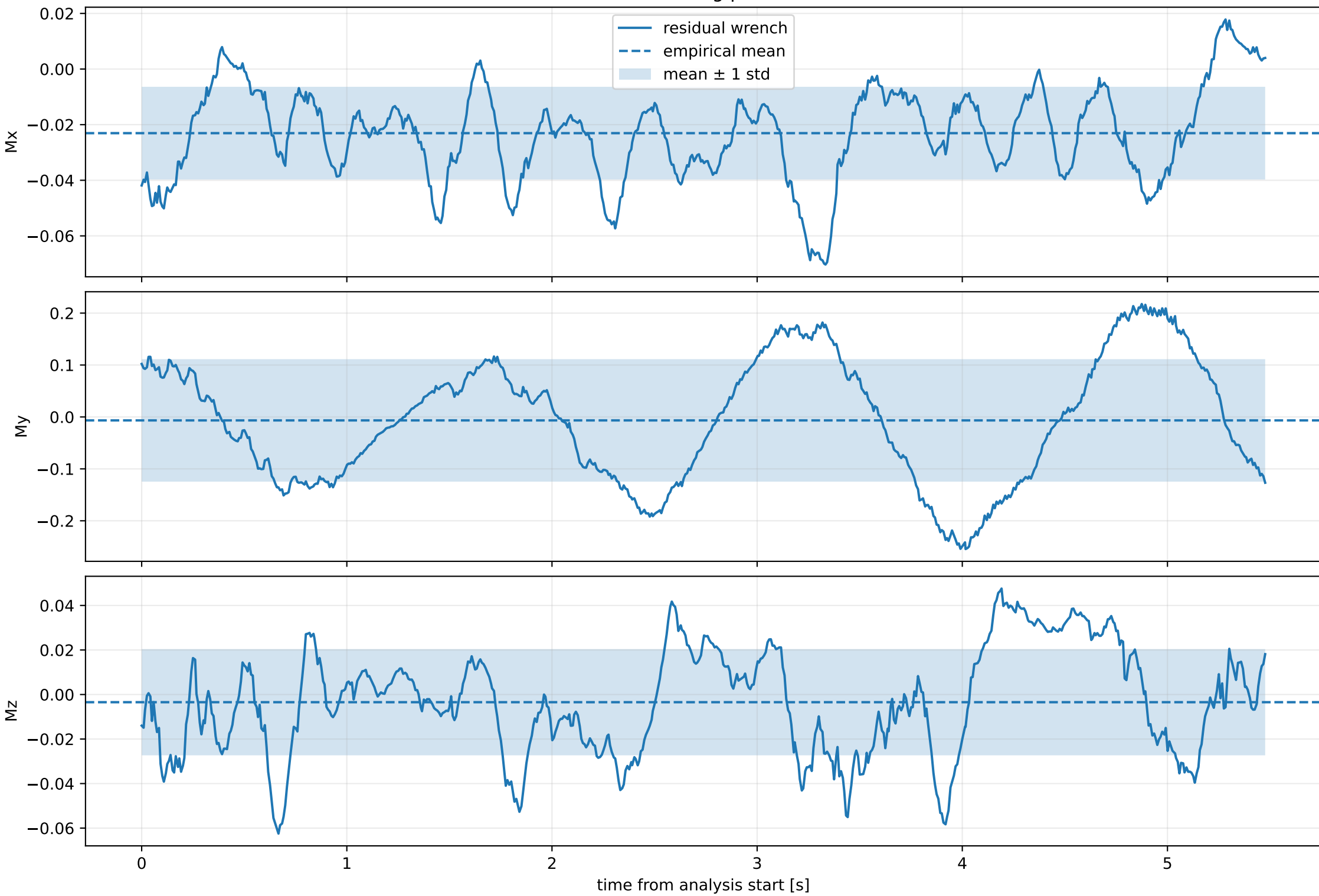
Parameter-only open-loop rollout (separate diagnostic)

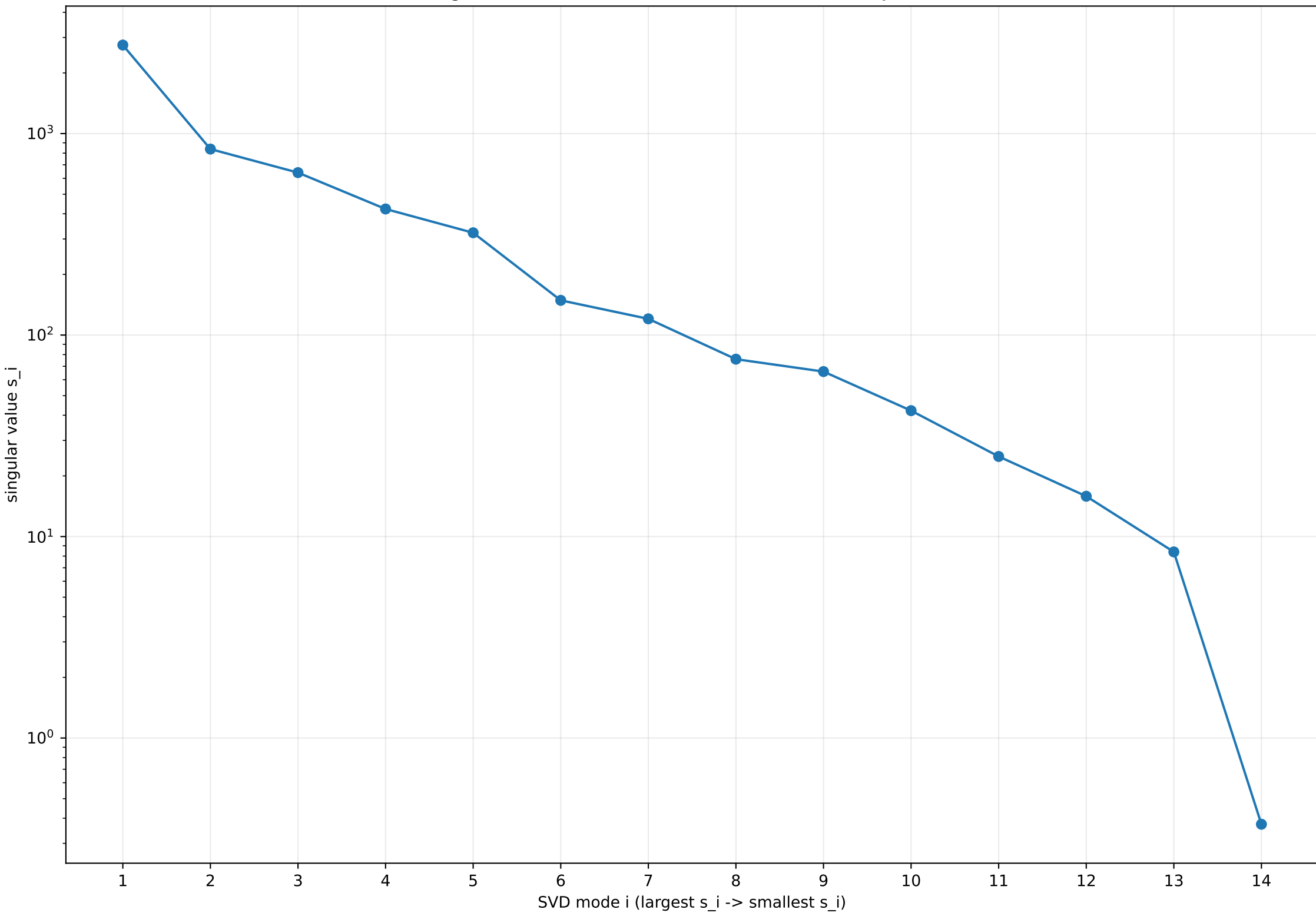


Dimensionless residual force: empirical mean $\pm$ 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence

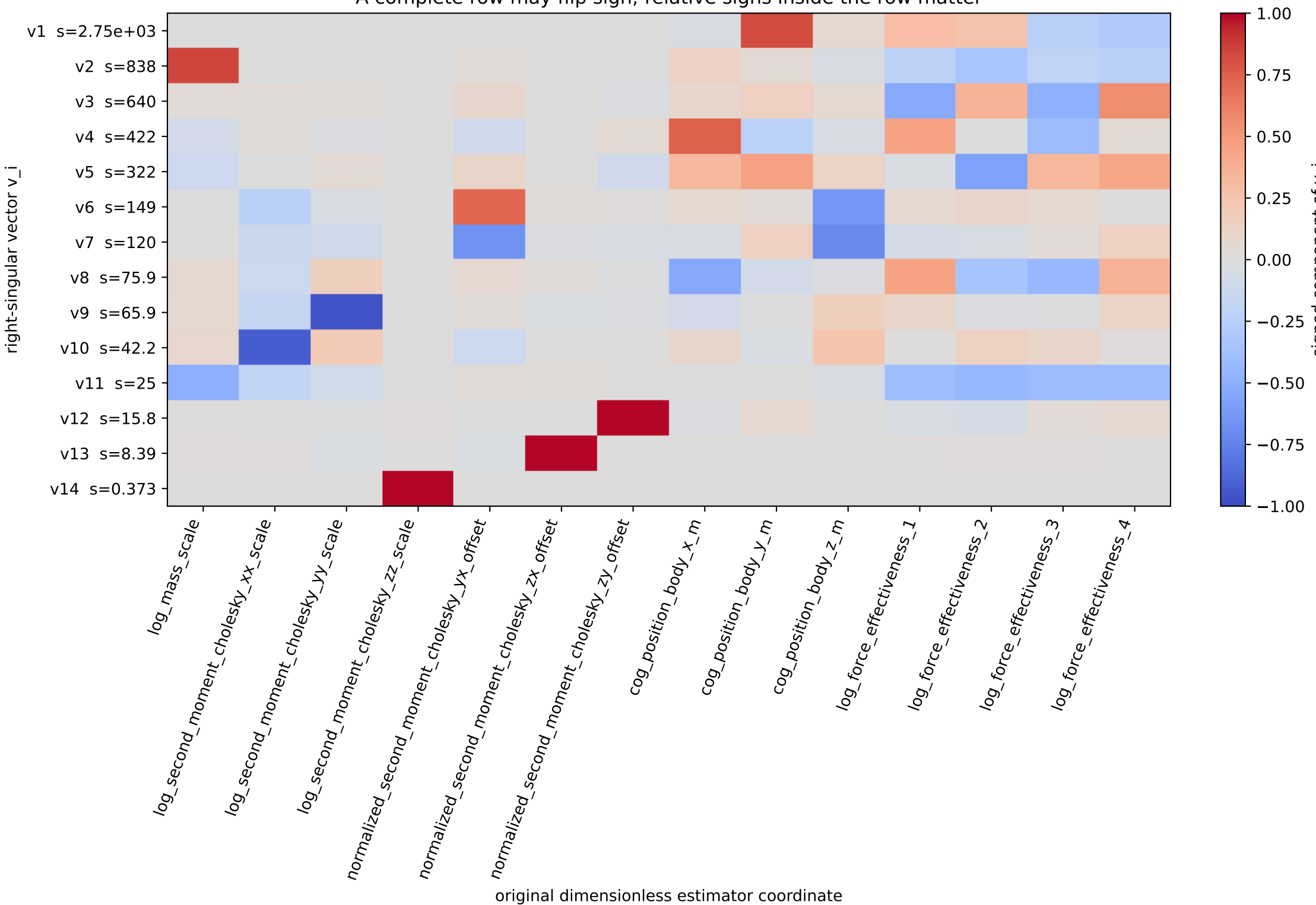


Dimensionless residual torque: empirical mean ± 1 std
Wide band = disturbance varies more, so matching parameter directions receive weaker evidence



Singular values S of the whitened dimensionless Jacobian

Signed right-singular vectors V^T
A complete row may flip sign; relative signs inside the row matter



Decoded physical parameters at the selected estimate

mass [kg]

2.35159759125

inertia J [kg m²]

[0.065000061 -7.2790028e-07 1.901508e-05]
[-7.2790028e-07 0.064952656 5.9167879e-08]
[1.901508e-05 5.9167879e-08 0.12899211]

principal moments [kg m²]

[0.064952645, 0.065000067, 0.12899212]

CoG position in body frame [m]

[-0.0020247086, -3.0526551e-05, 0.0095097496]

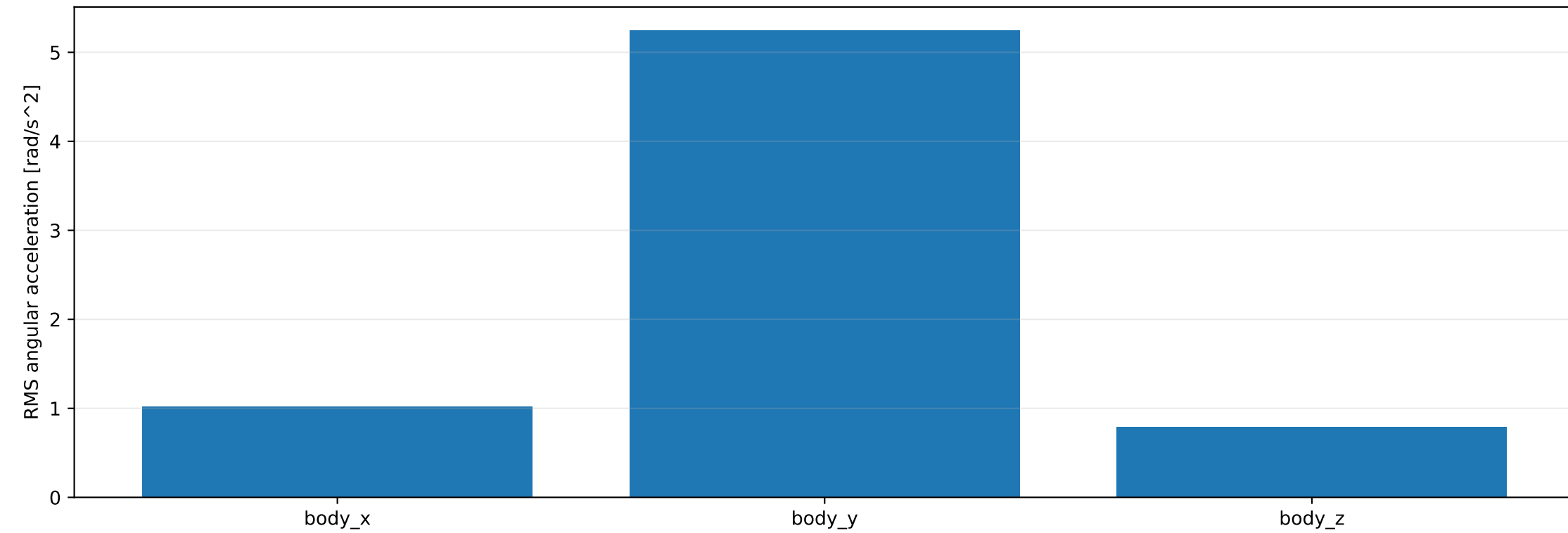
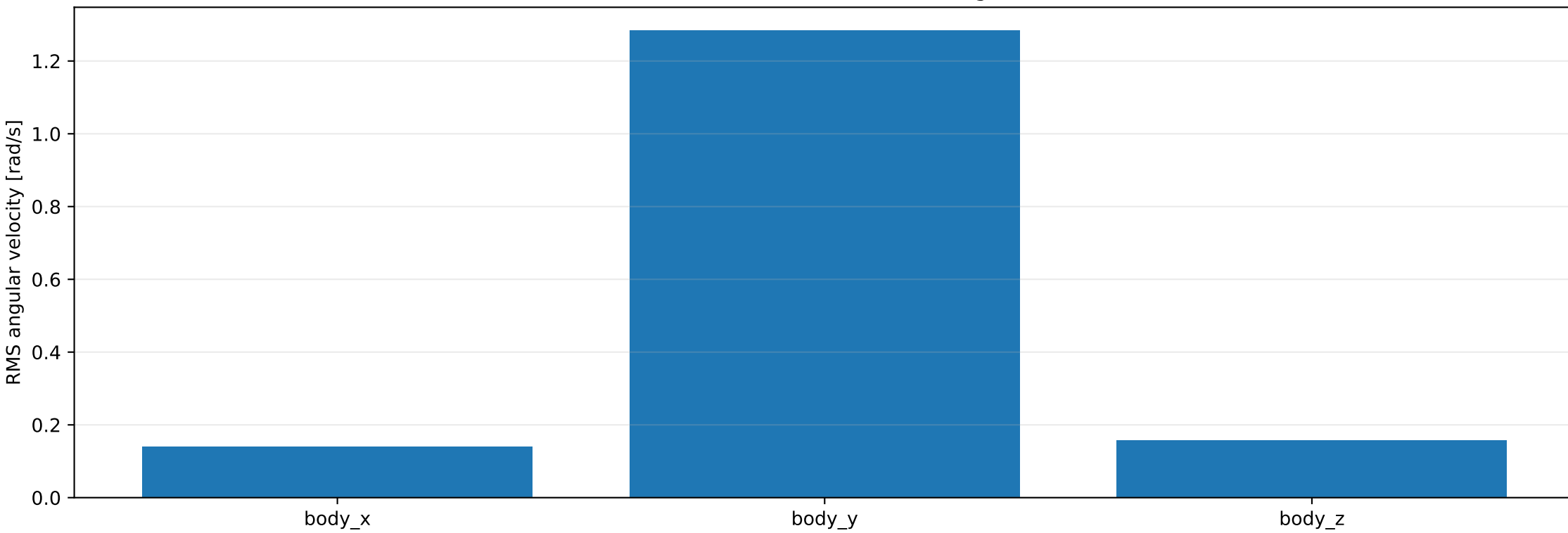
rotor force effectiveness

[1, 1, 1, 1]

selected delay [s]

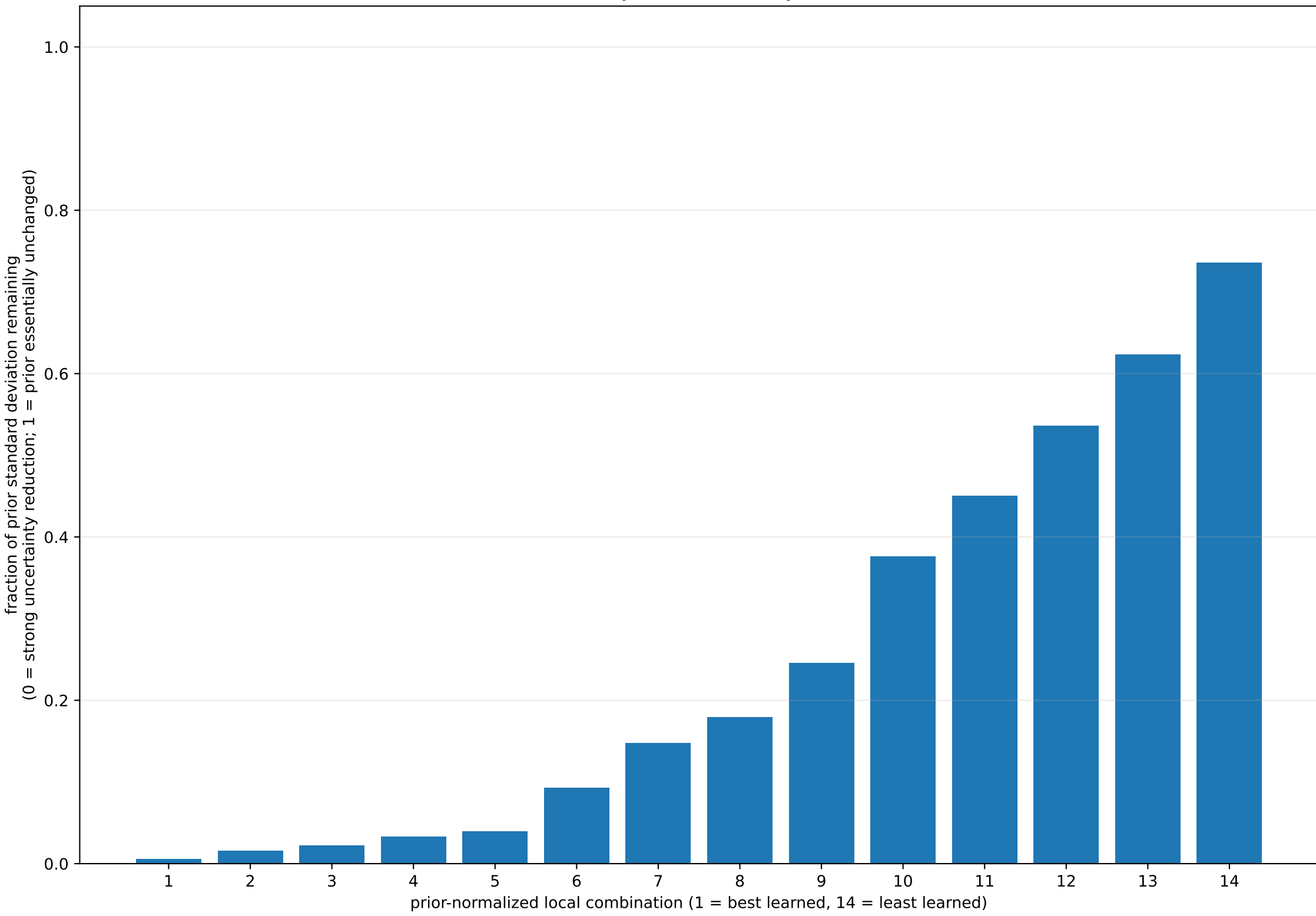
0.004000000392

Axis-wise rotational excitation diagnostic



Low body-z values can support a yaw-excitation hypothesis, but identifiability is decided by the full Jacobian.

How much uncertainty remains in every local combination?



Confidence accumulation as the rosbag becomes longer

